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import pandas as pd
import numpy as np
from faker import Faker
import random

fake = Faker()

# Define the number of records
num_records = 3000

# Project managers
project_managers = [fake.name() for _ in range(10)]

# Team members
team_members = [fake.name() for _ in range(300)]

# Possible project types and statuses
project_types = ['Development', 'Research', 'Testing', 'Deployment', 'Maintenance']
project_statuses = ['Not Started', 'In Progress', 'Completed', 'On Hold', 'Cancelled']

data = []

# Sets to store unique project names and IDs
project_names_set = set()
project_ids_set = set()

# pre-defined project health values
health_needs_attention = 50
health_not_set = 11
health_on_track = num_records - health_needs_attention - health_not_set

#pre-defiend priorety values
priority_weights = {
    'Low': 0.4,
    'Medium': 0.35,
    'High': 0.2,
    'Critical': 0.05
}

#generate the project health list
project_health_list = (['Needs Attention'] * health_needs_attention +
    ['Not Set'] * health_not_set +
    ['On Track'] * health_on_track)

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random.shuffle(project_health_list)

for i in range(num_records):
    #generate a unique project ID
    while True:
        project_id = fake.uuid4()
        if project_id not in project_ids_set:
            project_ids_set.add(project_id)
            break

    #Generate a unique project name
    while True:
        project_name = fake.bs().title()
        if project_name not in project_names_set:
            project_names_set.add(project_name)
            break

    project_manager = random.choice(project_managers)
    num_team_members = random.randint(5, 15)
    assigned_team_members = random.sample(team_members, num_team_members)
    start_date = fake.date_between(start_date='-2y', end_date='today')
    end_date = fake.date_between(start_date=start_date, end_date='+1y')
    budget = round(random.uniform(10000, 99999), 2)
    expenses = round(random.uniform(2350, 100000), 2) #allowing some of expnses to go over
the budget

    #choose project health from the pre-defined list
    health = project_health_list[i]

    #choose priority based on weight distrubution
    priority = random.choices(
        population=list(priority_weights.keys()),
        weights=list(priority_weights.values()),
        k=1
    )[0]

    status = random.choice(project_statuses)
    project_type = random.choice(project_types)
    deployment_date = fake.date_between(start_date=start_date, end_date=end_date)

    data.append([
        project_id, project_name, project_manager, ', '.join(assigned_team_members), start_date,
        end_date, budget, expenses, priority, status, health, project_type, deployment_date

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)
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#create DataFrame
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columns = [  
    'Project ID', 'Project Name', 'Project Manager', 'Team Members', 'Start Date', 'End Date',  
    'Budget', 'Expenses', 'Priority', 'Project Status', 'Project Health', 'Project Type', 'Deployment  
Date'  
]  
df = pd.DataFrame(data, columns=columns)
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#Save to CSV
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df.to_csv('project_managment_dataset.csv', index=False)
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print("Dataset created successfully with the following fields:")
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print(df.head())
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